

Zhihan Zhou

Foundation Model Researcher | Research Scientist, NVIDIA

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Research Profile

Foundation model researcher worked on the full lifecycle—**data**, **large-scale pre-training**, **evaluation**, and **post-training**, at the intersection of **biological sequence modeling** and **scientific machine learning**. Developed the **DNABERT family** of genome foundation models, reaching **15M+ Hugging Face downloads** and **1,000+ GitHub stars**. Built the end-to-end training, inference, evaluation, and post-training pipeline for **GenomeOcean**, a **25B-parameter genome foundation model** trained from scratch for biological sequence understanding and protein-coding gene generation. Currently developing **MambaSV**, a GPU-native deep sequence model and simulator for haplotype-resolved germline and somatic structural variant calling from long-read sequencing.

Education

Northwestern University <i>Ph.D., Computer Science</i>	2020 – 2025 <i>Advisor: Han Liu</i>
Northwestern University <i>M.S., Computer Science</i>	2018 – 2020
Zhejiang University <i>B.S., Mathematics and Applied Mathematics</i>	2014 – 2018

Professional Experience

NVIDIA <i>Research Scientist</i>	03/2025 – Present <i>AI for Genomics</i>
- Developed MambaSV, a GPU-native deep sequence model for haplotype-resolved germline and somatic structural variant calling from matched tumor-normal long-read BAM files. - Designed a scalable paired normal-tumor genome simulator with accurate haplotype-aware SV labels, modeling tumor purity, low variant allele fraction, copy-neutral loss of heterozygosity, SNP/INDEL background, PacBio/ONT read statistics, and complex BND/fusion breakpoint signatures. - Achieved state-of-the-art performance across simulated and real long-read datasets for germline and somatic SV calling.	
Snap Inc. <i>Research Intern, User Modeling and Personalization</i>	05/2023 – 09/2023 <i>Foundation models for large-scale user representation</i>
- Developed foundation models to learn general-purpose user embeddings from behavioral logs for millions of users, covering data representation, model design, training objectives, and evaluation pipelines. - Research led to two papers and a production-oriented system for large-scale user representation learning.	
Amazon AWS AI Labs <i>Applied Scientist Intern</i>	09/2021 – 12/2021 <i>Large-scale contrastive pretraining for dialogue understanding</i>
Improved few-shot dialogue understanding using large-scale contrastive conversational model pretraining; findings and models were published at <i>NAACL 2022</i>.	

Selected Research Projects

DNABERT Family – Widely adopted genome foundation models	2020 – 2024
- Developed DNABERT, DNABERT-2, and DNABERT-S, a family of pretrained genome foundation models for DNA sequence modeling, multi-species genome representation learning, and species-aware biological embedding. - Built pre-training pipelines, post-training recipe, and evaluation workflows for biological sequence foundation models, bridging NLP-style language modeling with genomic data. - Achieved state-of-the-art performance in a wide range of genomics problems, including specials differentiation, metagenomics binning, functional elements prediction, etc. - Received broad open-source adoption with 15M+ Hugging Face downloads and 1,000+ GitHub stars; DNABERT is among the most widely used genome foundation models.	
GenomeOcean – Large-scale genome foundation model pretraining	2024 – 2025
- Built the entire end-to-end training and inference pipeline for a 25B-parameter genome foundation model trained from scratch, including multi-stage data curation, training-data design, context-length scheduling, distributed training orchestration, and evaluation. - Collaborated with Lawrence Berkeley National Laboratory and the Joint Genome Institute to build a large and diverse genome pretraining corpus from metagenomic assemblies for scalable foundation-model training.	

- Designed multi-stage pretraining across heterogeneous genomic datasets and varying context lengths to improve genome understanding and microbial design.
- GenomeOcean is 10X faster than baselines with similar size, and it is able to generate protein-coding genes constrained by evolutionary principles at a comparable level of strongest baseline (e.g., Evo2).

Selected Publications

- **Zhihan Zhou**, et al. *MambaSV: Accurate germline and somatic structural variant calling from long-reads with deep sequence model*. Work in progress. Presented in AGBT 2026 and AACR 2026.
- **Zhihan Zhou**, Robert Riley, Satria Kautsar, Weimin Wu, Rob Egan, Steven Hofmeyr, Shira Goldhaber-Gordon, Mutian Yu, Harrison Ho, Fengchen Liu, Feng Chen, Rachael Morgan-Kiss, Lizhen Shi, Han Liu, Zhong Wang. *GenomeOcean: An Efficient Genome Foundation Model Trained on Large-Scale Metagenomic Assemblies*. Under review at Nature Methods.
- Yanyong Ji*, **Zhihan Zhou***, Han Liu, Ramana V. Davuluri. *DNABERT: Pre-trained Bidirectional Encoder Representation from Transformers for DNA Language Modeling in Genome*. Bioinformatics, 2021. (* co-first author)
- **Zhihan Zhou**, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, Han Liu. *DNABERT-2: Efficient Foundation Model and Benchmark for Multi-Species Genome*. ICLR, 2024.
- **Zhihan Zhou***, Weimin Wu*, Harrison Ho, Lizhen Shi, Zhong Wang, Han Liu. *DNABERT-S: Learning Species-Aware DNA Embedding with Genome Foundation Model*. ISMB 2025. (* co-first author)
- Qixiang Fang*, **Zhihan Zhou***, Leonardo Neves, Francesco Barbieri, Yozen Liu, Dong Nguyen, Daniel L. Oberski, Maarten W. Bos, Ron Dotsch. *General-Purpose User Modeling with Behavioral Logs: A Case Study with Snapchat*. SIGIR, 2024. (* co-first author)
- **Zhihan Zhou**, Dejiao Zhang, Wei Xiao, Nicholas Dingwall, Xiaofei Ma, Andrew Arnold, Bing Xiang. *Learning Dialogue Representations from Consecutive Utterances*. NAACL, 2022.
- Chunyu Ma, **Zhihan Zhou**, Han Liu, David Koslicki. *KGML-xDTD: A Knowledge Graph-based Machine Learning Framework for Drug Treatment Prediction and Mechanism Description*. GigaScience, 2023.

Technical Skills

Foundation model	Data (collection, curation and simulation), pre-training (generative and masked), post-training (contrastive, SFT, and RLHF), deployment (inference and vectorized database) and evaluation (verifiable and human-in-the-loop).
ML systems	Distributed training, multi-stage pre-training recipe, data curation and streaming, Docker/Kubernetes, HPC workflows.
Scientific ML	Genomics, long-read sequencing, structural variants, cell-free DNA, knowledge graphs.

Invited Talks & Professional Service

- **Invited talks:** NVIDIA, NIH, Berkeley Lab, Harvard Medical School, Stony Brook University; topics include the DNABERT family, species-aware genome foundation models, and the future of genome foundation models.
- **Reviewer:** NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, AISTATS, and related machine learning / NLP venues.
- **Teaching:** Teaching assistant for courses in artificial intelligence, statistical language modeling, deep learning, machine learning, NLP, and data science.